CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

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1. INTRODUCTION

1.1 Project Overview

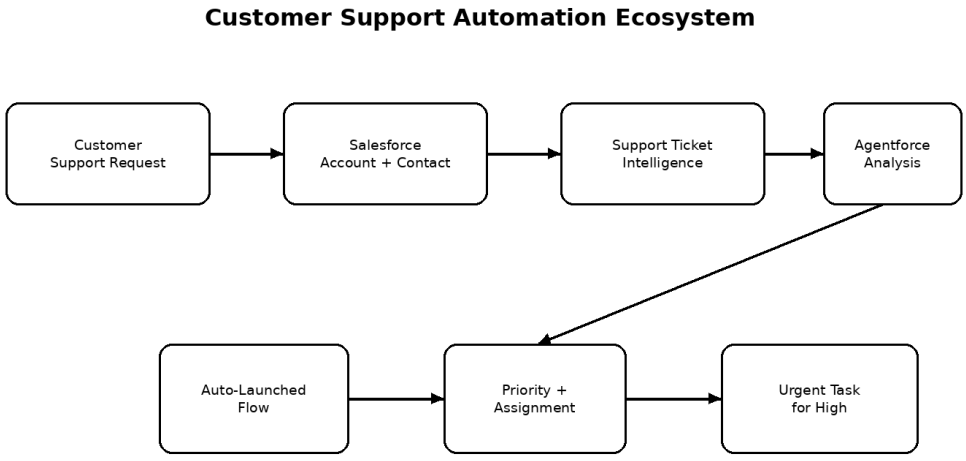
Customer support teams receive a large number of requests and must identify which cases need immediate attention. This project implements a Salesforce-based workflow that retrieves the latest ticket for a customer Account, analyzes its description, determines a High, Medium or Low priority, and returns the handling result through Agentforce.

1.2 Purpose

- Reduce repetitive manual ticket prioritization.
- Provide a conversational interface through Agentforce.
- Use Salesforce Flow as the controlled backend automation layer.
- Create an urgent Task automatically when a High-priority condition is detected.
- Return a clear priority, assigned support level and action message.

1.3 Project Objectives

Objective	Expected Result
Ticket identification	Retrieve the latest ticket linked to the requested Account.
Priority classification	Classify the description into High, Medium or Low using configured conditions.
Automated handling	Create an urgent Task for High-priority cases.
Support assignment	Return an appropriate support level such as Senior Support Agent.
Conversational output	Present the result through Agentforce.



2. IDEATION PHASE

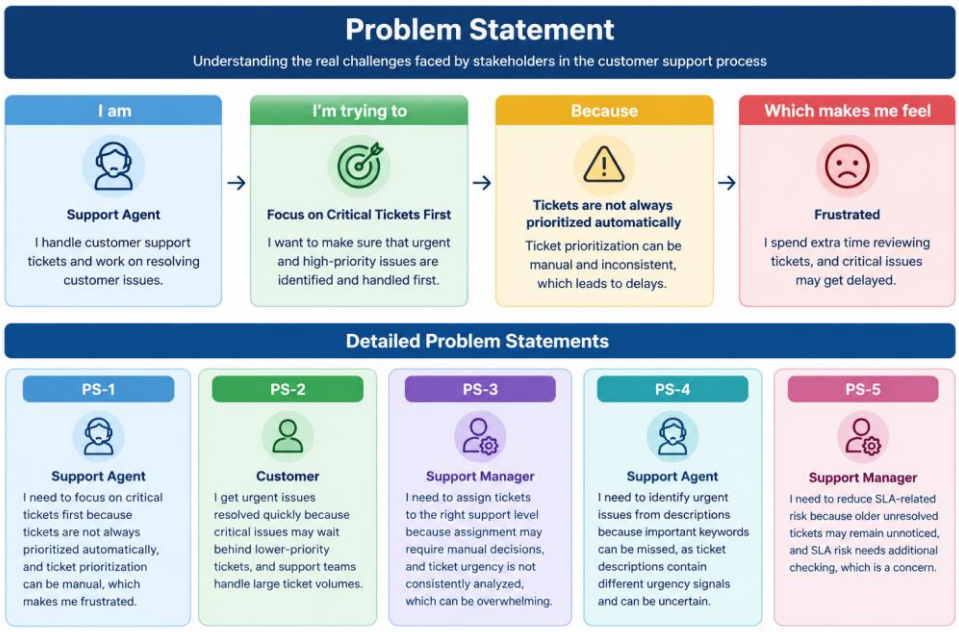
2.1 Problem Statement

Support teams may spend time reading ticket descriptions, deciding urgency and deciding which level of support should handle a case. When ticket volume increases, important urgency signals can be missed or handled inconsistently.

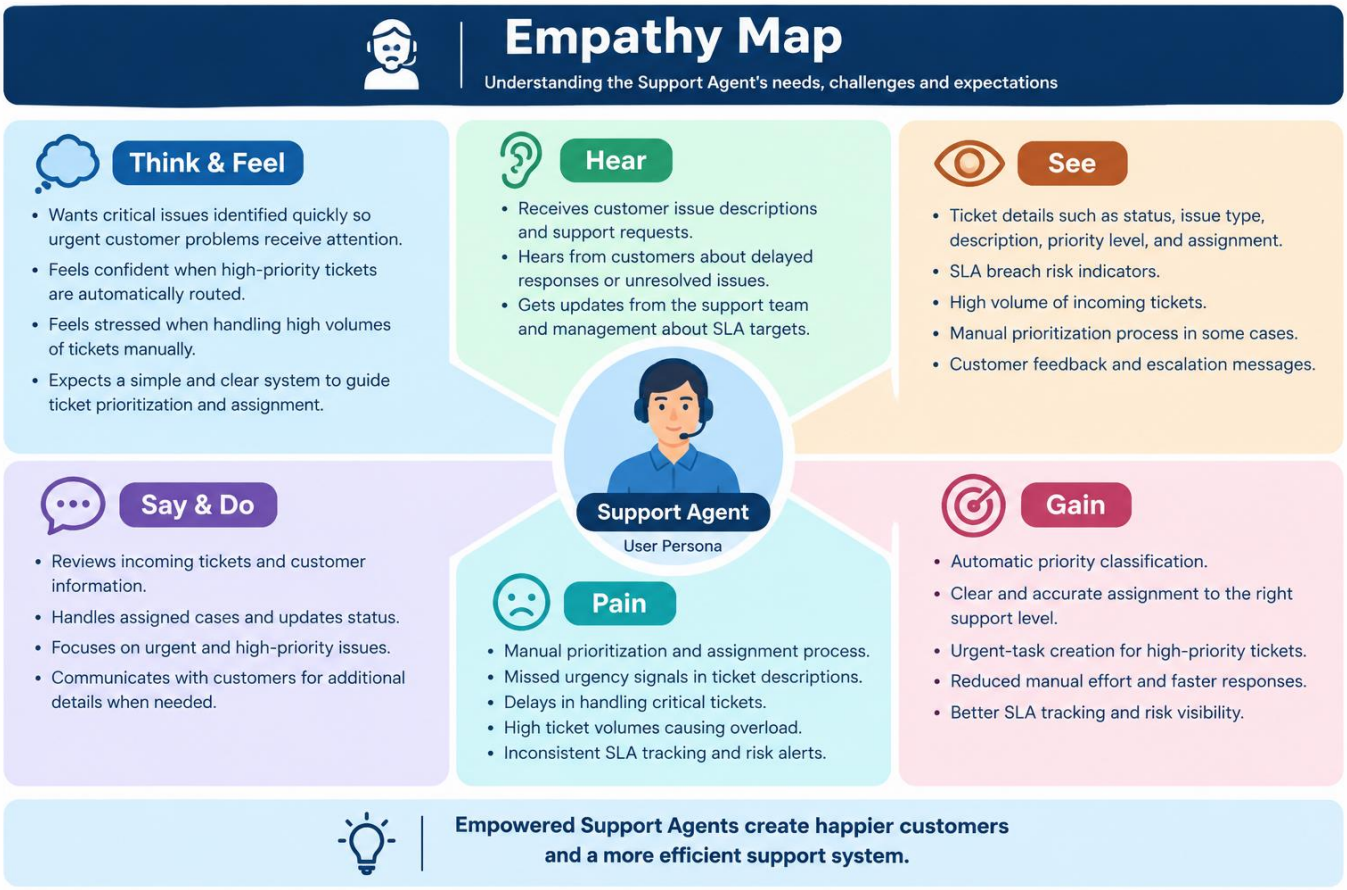
Stakeholder	Need	Current Difficulty	System Response
Support Agent	See critical tickets first	Manual review of descriptions	Automatic priority classification
Customer	Receive quick attention for urgent issues	Critical issues can wait behind other tickets	High-priority handling path
Support Manager	Route work consistently	Assignment decisions can be manual	Configured support-level output
Agentforce	Analyze ticket information	Needs controlled backend action	Agentforce invokes Flow

2.2 User Personas

Persona	Main Goal	Interaction With System
Support Agent	Resolve tickets efficiently	Receives priority and handling information
Customer	Get urgent issues handled quickly	Creates support request
Support Manager	Maintain consistent support operations	Monitors assignment and SLA-related risk
Agentforce	Interpret the request and initiate automation	Collects Account Name and invokes Flow



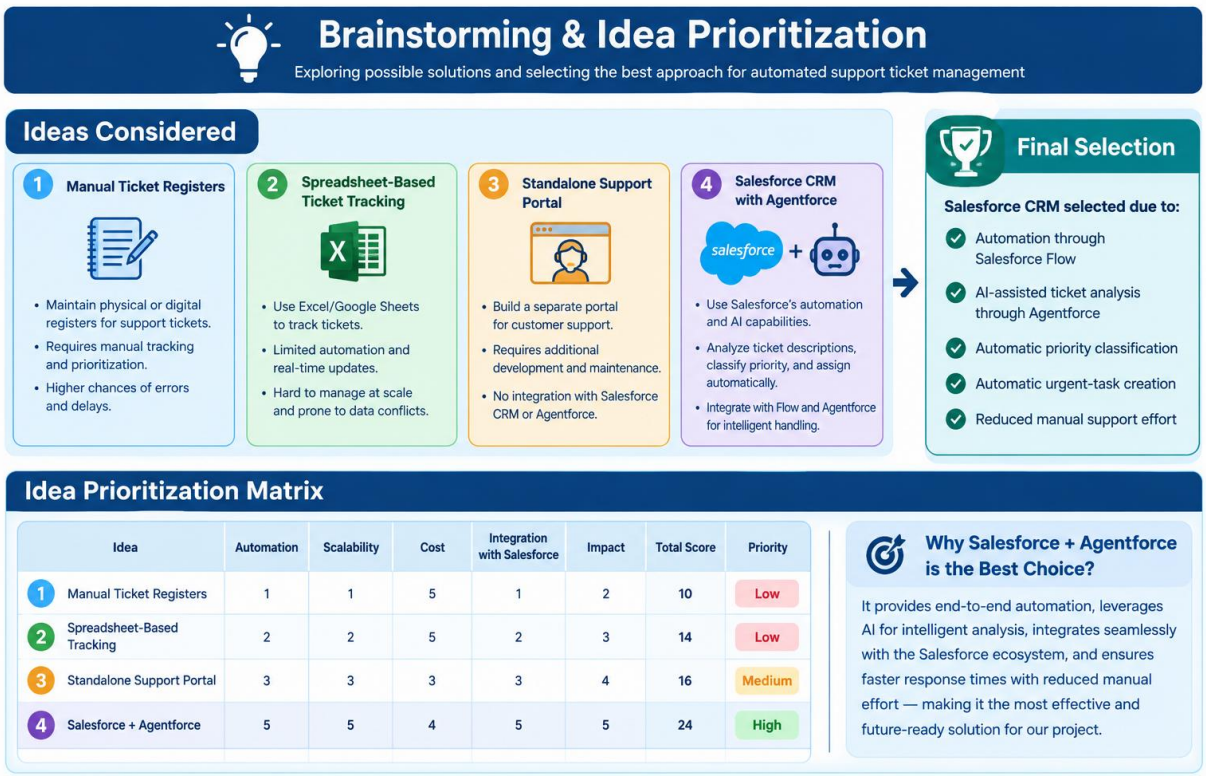
2.3 Empathy Map Canvas



2.4 Brainstorming and Idea Selection

The ideation considered progressively more automated approaches. The selected direction combines CRM data, deterministic backend automation and a conversational AI interface.

- Manual ticket register – simple but highly dependent on human effort.
- Spreadsheet tracking – easy to start but weak for controlled automation.
- Standalone portal – improves interface but requires a separate support platform.
- Salesforce Flow – provides structured backend automation.
- Salesforce + Agentforce – combines CRM data, automation and conversational interaction.



3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map



The journey moves from a customer issue to structured Salesforce data, automated analysis and support handling. The key automation checkpoints are ticket retrieval, description analysis, priority classification, assignment and urgent-task creation.

3.2 Functional Requirements

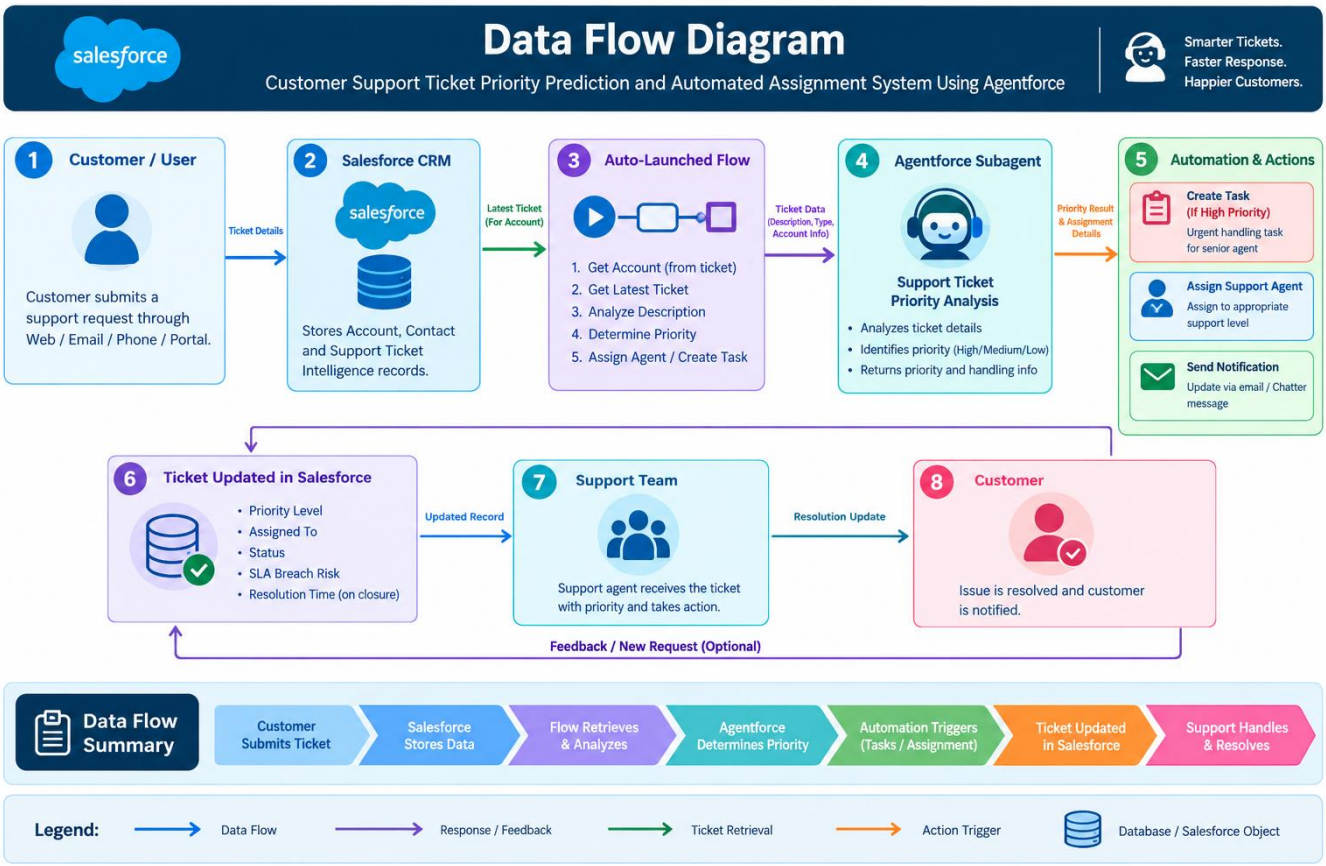
ID	Requirement	Description
FR-01	Ticket Management	Create and maintain Support Ticket Intelligence records.
FR-02	Account Association	Associate the ticket with an Account and optionally a Contact.
FR-03	Latest Ticket Retrieval	Find the most recent ticket for the requested Account.
FR-04	Description Analysis	Evaluate configured urgency indicators in the description.
FR-05	Priority Classification	Return High, Medium or Low.
FR-06	High-Priority Task	Create Urgent Ticket Handling Task for High priority.
FR-07	Support Assignment	Return the configured support level.
FR-08	Agentforce Interaction	Allow the user to request analysis conversationally.
FR-09	SLA Risk Check	Provide an optional check for older unresolved tickets.

3.3 Non-Functional Requirements

Quality	Requirement
Usability	Agentforce prompts should be understandable to support users.
Reliability	The same configured conditions should produce consistent results.
Performance	Latest-ticket retrieval and Flow execution should complete quickly.
Security	Customer and ticket information should remain within controlled Salesforce access.
Maintainability	Priority rules and Flow actions should be easy to update.
Scalability	The design should support more tickets and future assignment rules.

3.4 Data Flow

User Account Name → Agentforce Subagent → Flow Input → Account Lookup → Latest Ticket Lookup → Description Analysis → Priority Output → Assignment / Task Action → Agentforce Response.



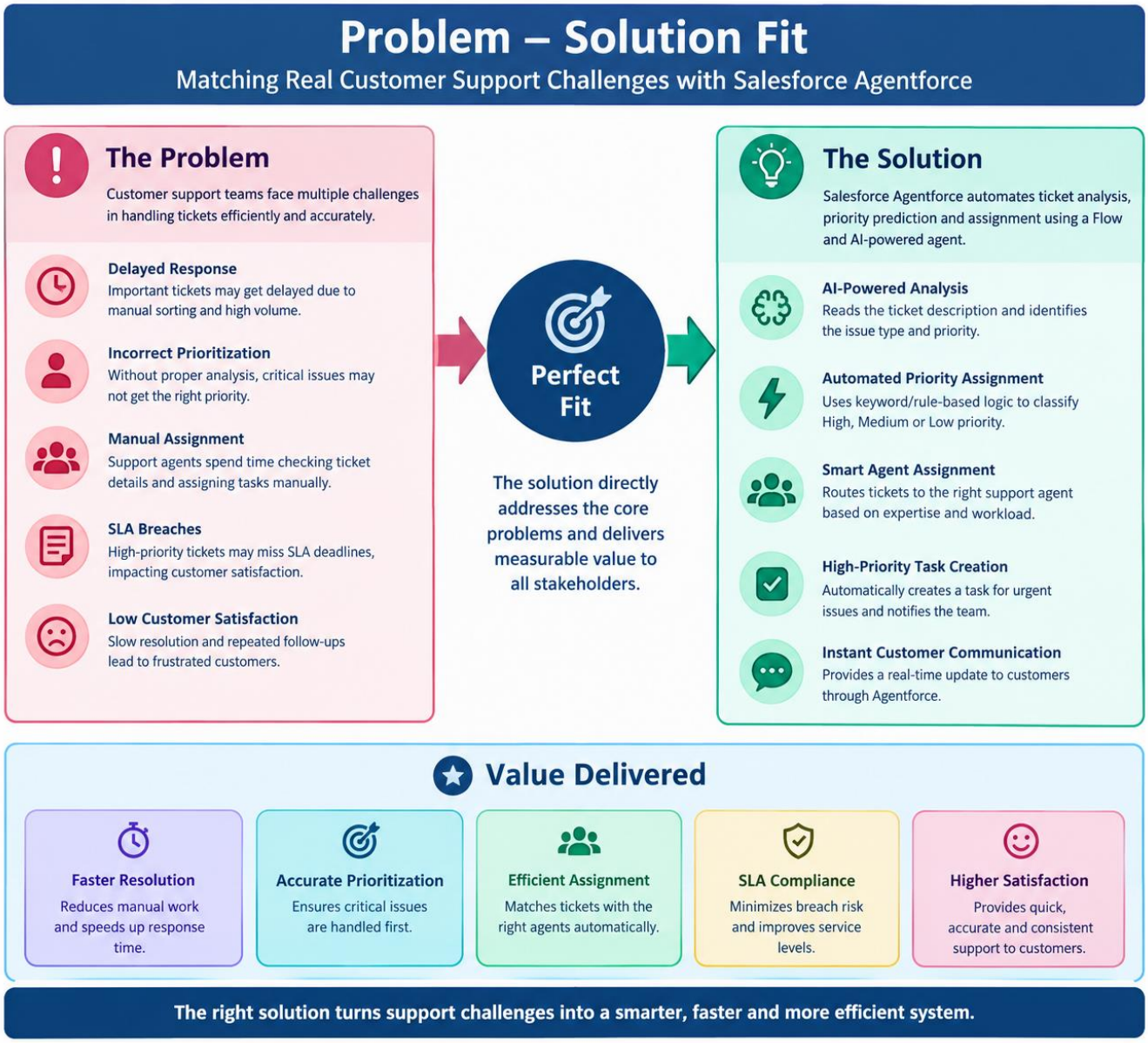
3.5 Technology Stack

Layer	Technology / Component
CRM Platform	Salesforce
Data	Support_Ticket_Intelligence__c, Account, Contact, User, Task
Automation	Auto-Launched Salesforce Flow
AI Interface	Agentforce
Agent Topic	Support Ticket Priority Analysis
Development / Deployment	Salesforce DX and Salesforce CLI
Version Control	Git and GitHub

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

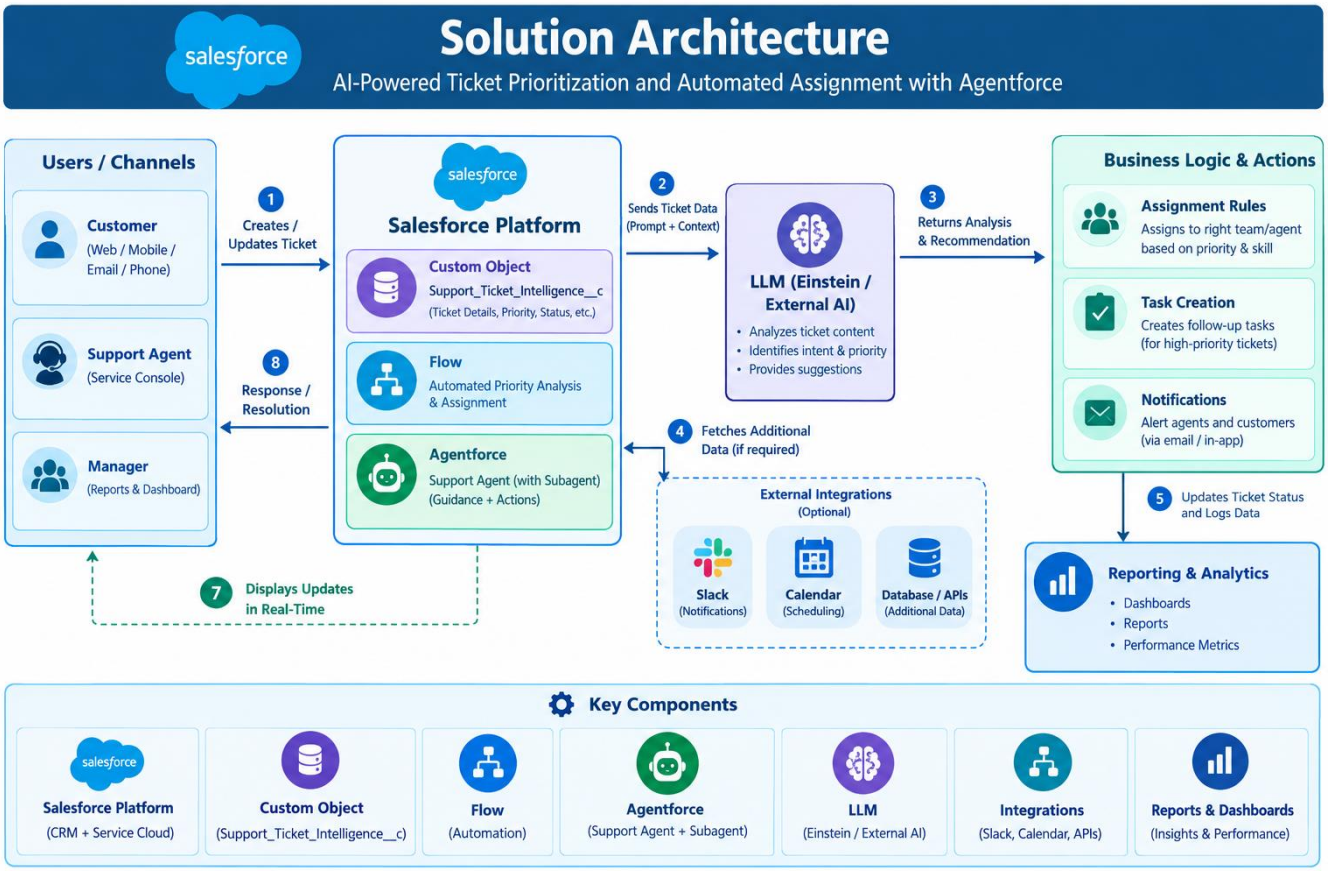
Problem Area	Designed Solution	Result
Manual prioritization	Description-based Flow decision	Priority returned automatically
Urgency can be missed	Explicit High-priority indicators	Urgent path is activated
Manual task creation	Flow creates Task	Less repetitive work
Manual support routing	Assigned support level output	Consistent handling rule
Conversational access	Agentforce subagent	User can request analysis



4.2 Proposed Solution

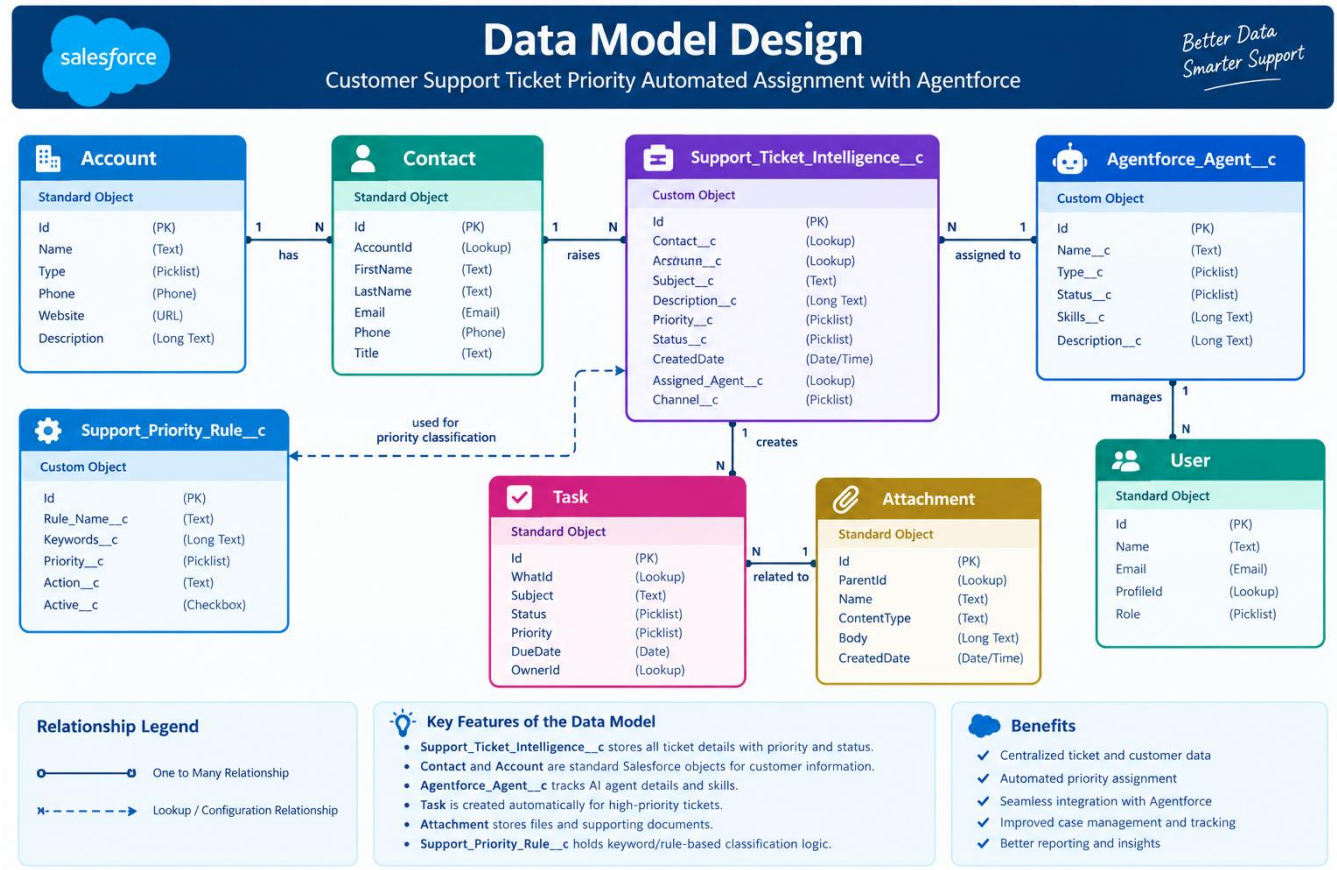
The proposed system combines a structured Salesforce custom object with an Auto-Launched Flow and an Agentforce subagent. Agentforce handles conversational interaction while the Flow performs record retrieval, priority decision logic, assignment output and Task creation.

4.3 Solution Architecture



4.4 Data Model Design

The Support Ticket Intelligence object is the central business record. It connects customer context from Account/Contact with issue information and operational fields such as priority, status, assignment and SLA risk. Task records are created for urgent handling.



4.5 Priority Decision Logic

The current implementation uses configured description keywords. High priority indicators are 'urgent', 'not working' and 'failure'. Medium indicators are 'issue', 'slow' and 'delay'. If none are detected, the default is Low.

5. PROJECT PLANNING & SCHEDULING

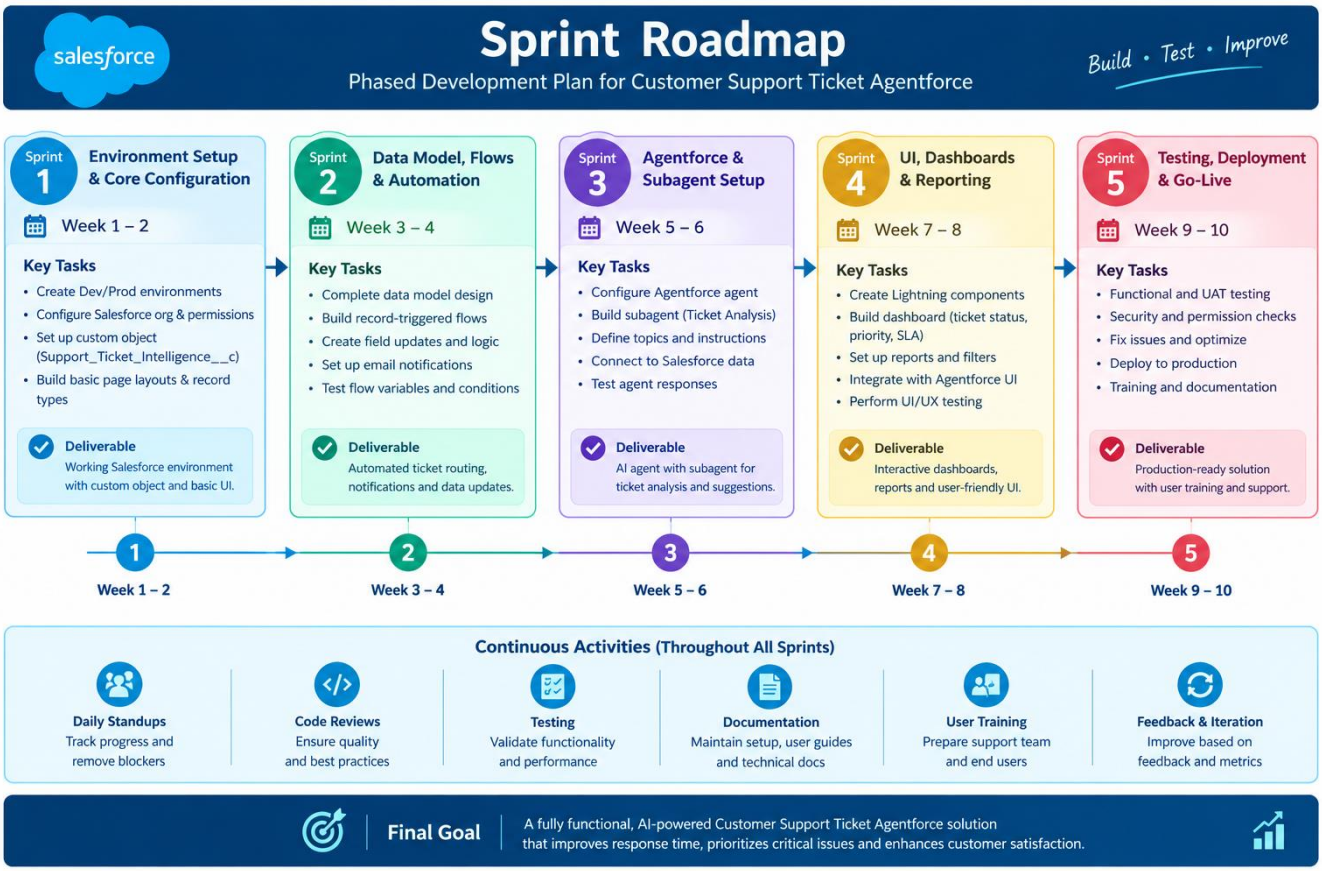
5.1 Agile Planning

- Break the project into small functional increments.
- Implement the data model before automation.
- Validate Flow behavior before connecting Agentforce.
- Test High, Medium and Low paths independently.
- Use Git to maintain the Salesforce project source.

5.2 Product Backlog

ID	Epic	User Story / Task	Points
US-01	Environment	Configure Salesforce project and development environment	3
US-02	Data Model	Create Support Ticket Intelligence custom object	5
US-03	Data Model	Create Account and Contact relationships	3
US-04	Automation	Build Auto-Launched Flow	5
US-05	Automation	Implement priority decision logic	5
US-06	Agentforce	Create Support Ticket Priority Analysis subagent	8
US-07	Assignment	Create High-priority Task and assignment output	5
US-08	SLA	Add optional SLA-risk checking	5
US-09	Testing	Validate Agentforce conversation and Flow outputs	5
US-10	Version Control	Retrieve Salesforce metadata and push to GitHub	3

5.3 Sprint Roadmap



5.4 Project Tracking

Sprint	Main Deliverable	Validation Point
Sprint 1	Salesforce environment	Org access and project initialized
Sprint 2	Custom object and fields	Records can be stored
Sprint 3	Flow automation	Priority and Task behavior verified
Sprint 4	Agentforce integration	Conversation invokes Flow
Sprint 5	Testing and refinement	High/Medium/Low scenarios validated

6. PROJECT DEVELOPMENT PHASE

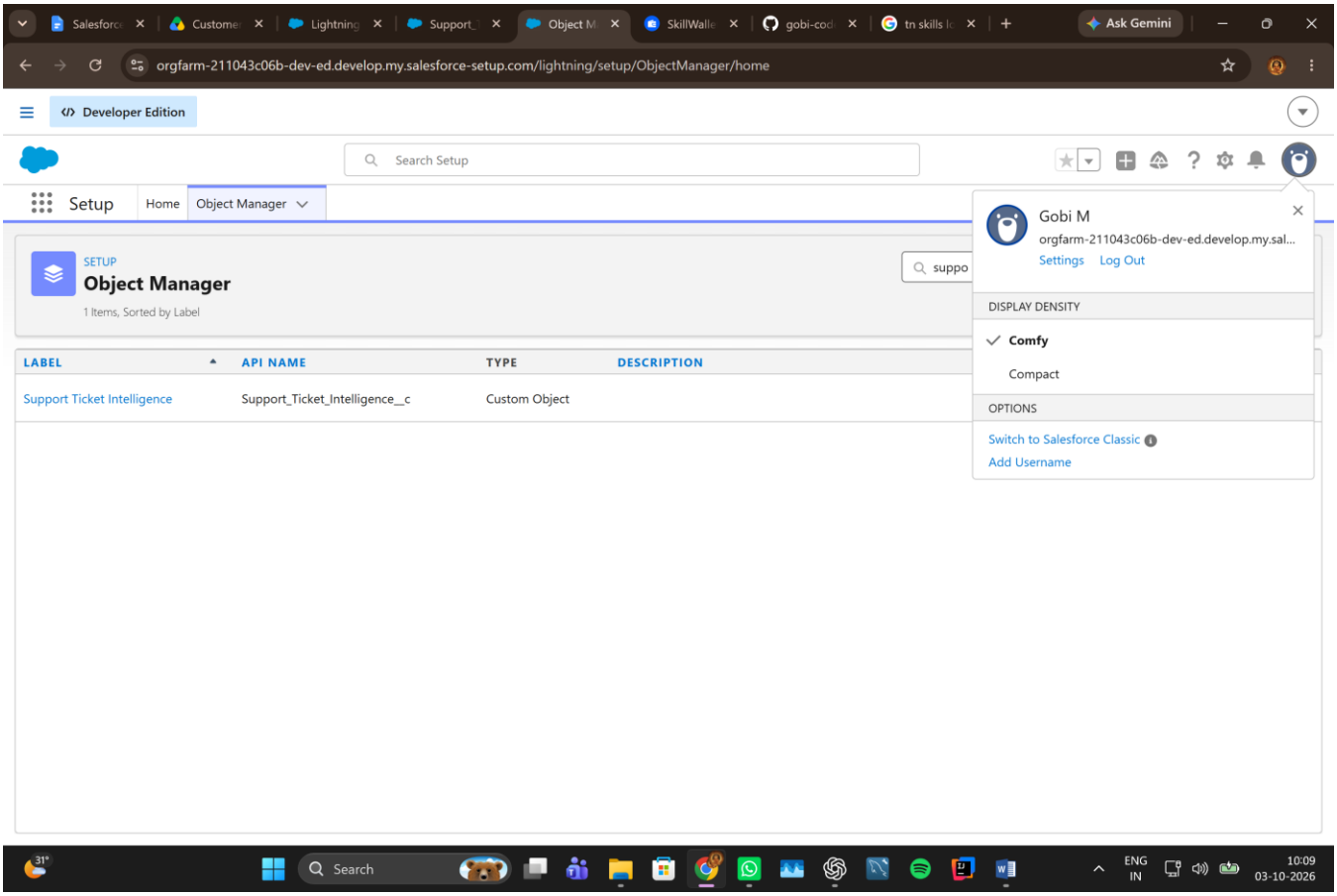
6.1 Salesforce Environment

The development environment uses Salesforce and a Salesforce DX project. The project source is maintained locally and synchronized with GitHub so metadata such as objects, flows and Agentforce configuration can be versioned.

Component	Project Value
Salesforce Org	Runtime configuration and testing
Salesforce DX Project	Local source representation
Salesforce CLI	Retrieve, inspect and deploy metadata
Git	Version control
GitHub	Remote project repository

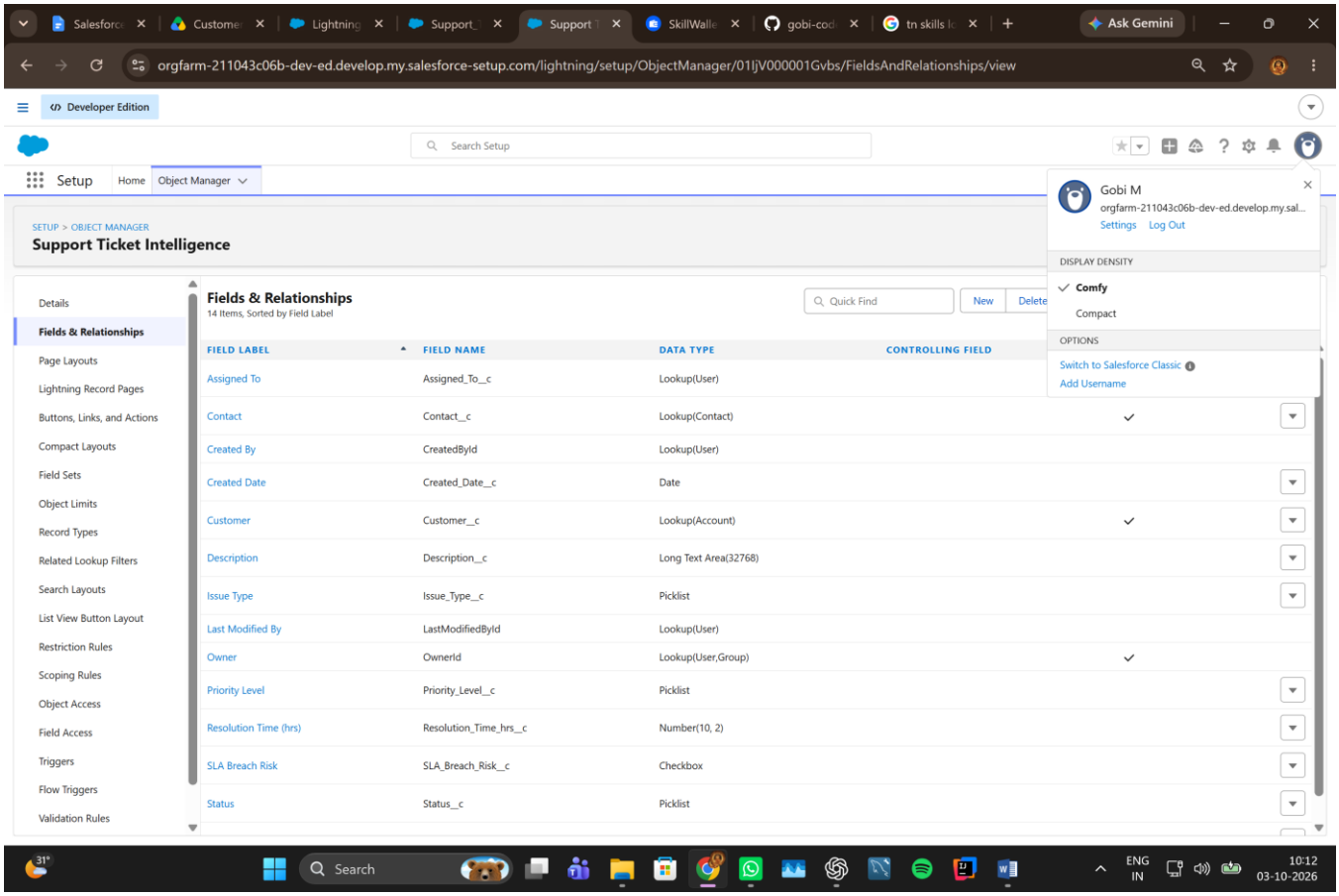
6.2 Custom Object

Property	Value
Label	Support Ticket Intelligence
API Name	Support_Ticket_Intelligence__c
Purpose	Stores customer support ticket information for analysis and automated handling



6.3 Fields and Relationships

Field	API Name	Type	Purpose
Ticket Number	Ticket_Number__c	Auto Number	Unique ticket identifier
Customer	Customer__c	Lookup(Account)	Links ticket to customer account
Contact	Contact__c	Lookup(Contact)	Links ticket to customer contact
Issue Type	Issue_Type__c	Picklist	Technical, Billing or General
Description	Description__c	Long Text Area	Stores issue details
Priority Level	Priority_Level__c	Picklist	Low, Medium or High
Status	Status__c	Picklist	New, In Progress or Resolved
Created Date	Created_Date__c	Date	Ticket date
Assigned To	Assigned_To__c	Lookup(User)	Support agent reference
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	Risk flag
Resolution Time	Resolution_Time_hrs__c	Number	Resolution duration in hours in the current org



6.4 Flow Design

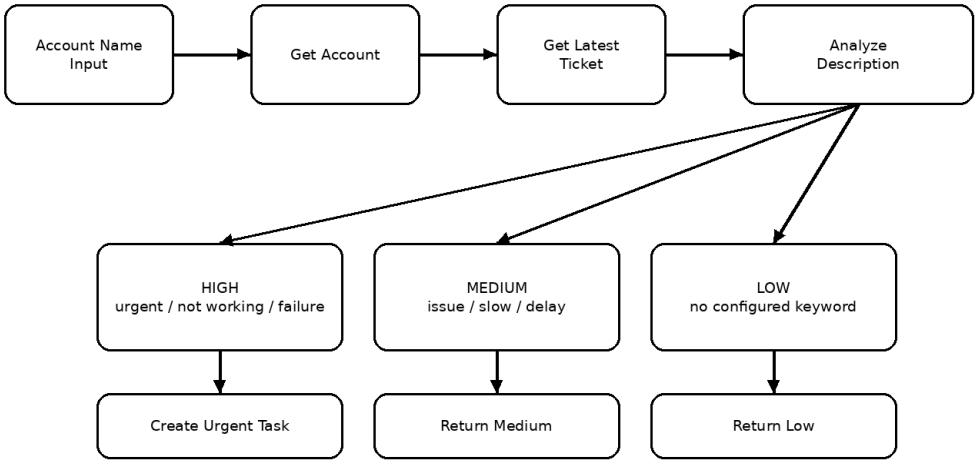
Flow type: Auto-Launched Flow. Input: Account Name. Outputs: Account Id, Ticket Id, Priority Level, Assigned Agent and Action Message.

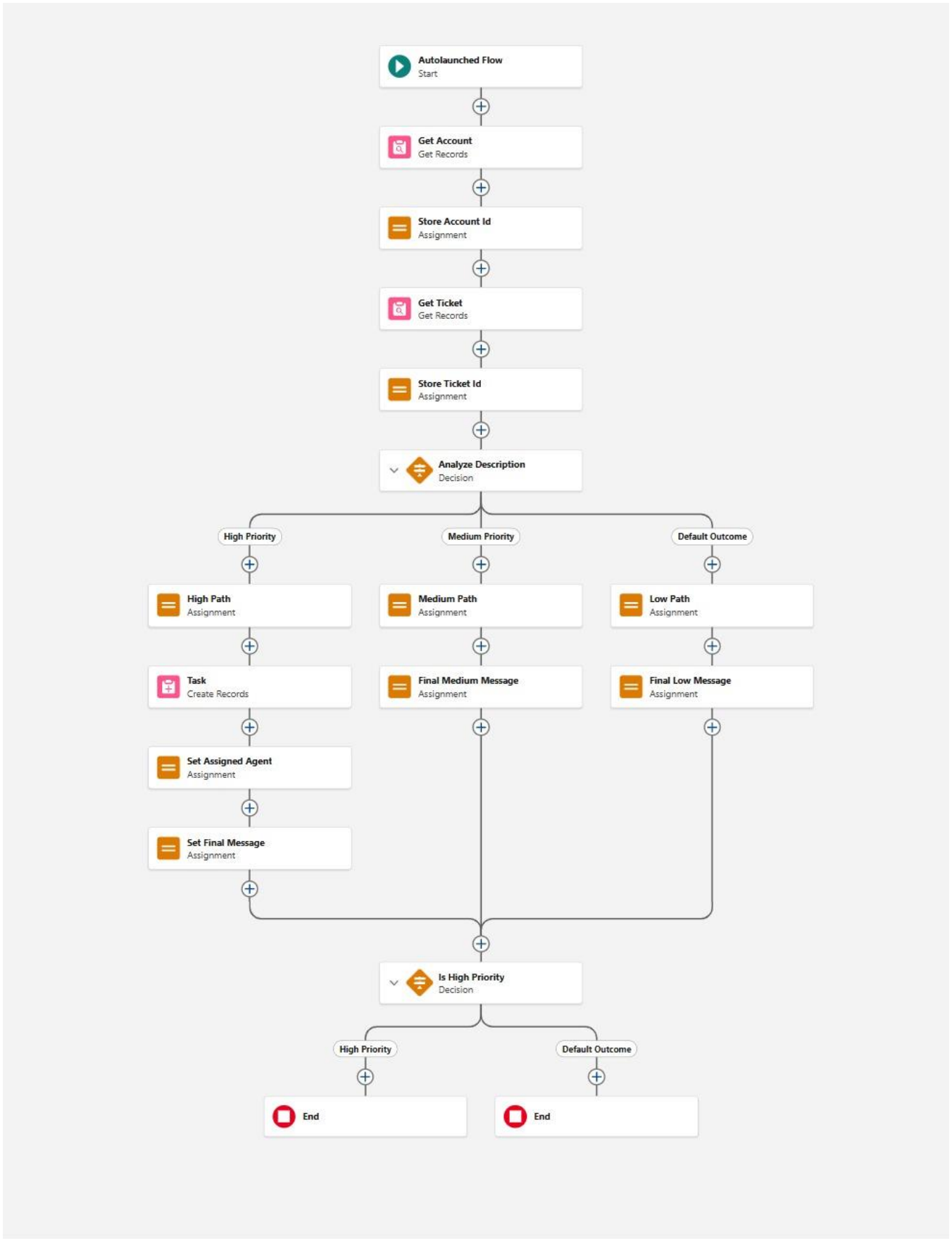
Variable	Type	Purpose
varAccountName	Text / Input	Account provided by Agentforce
varAccountId	Text / Output	Resolved Account Id
varTicketId	Text / Output	Latest Support Ticket Id
varPriorityLevel	Text / Output	High / Medium / Low
varAssignedTo	Text / Output	Support handling level
varActionMessage	Text / Output	Human-readable result

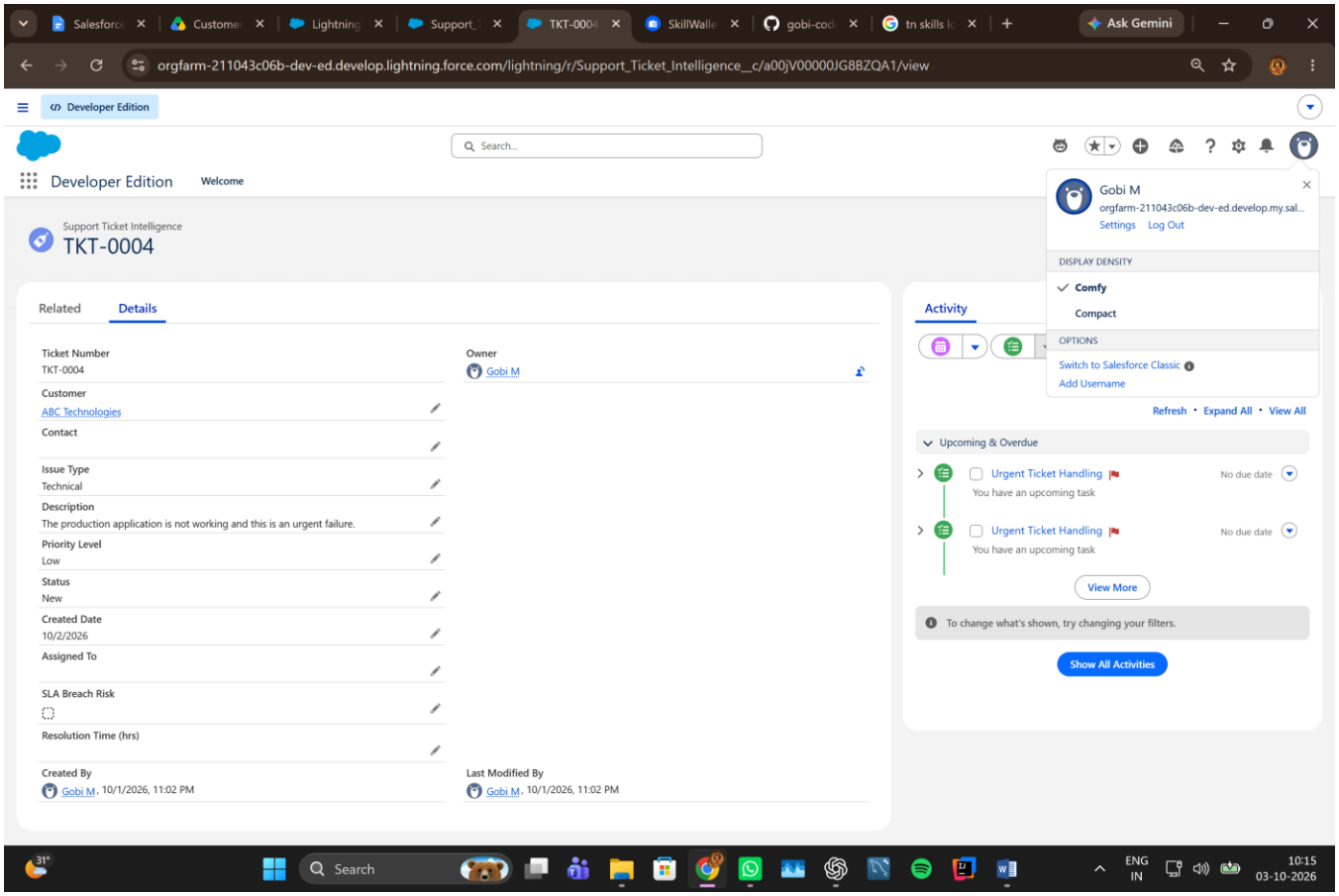
6.5 Flow Implementation

Stage	Implementation
1. Account Lookup	Get Account where Name equals varAccountName; latest record first.
2. Account Id	Store Get Account Id in varAccountId.
3. Ticket Lookup	Get latest Support Ticket where Customer__c equals varAccountId.
4. Ticket Id	Store Get Ticket Id in varTicketId.
5. Description Decision	Check configured High and Medium urgency indicators.
6. Priority Assignment	Set varPriorityLevel to High, Medium or Low.
7. High Handling	Create Urgent Ticket Handling Task for High priority.
8. Assignment	Set varAssignedTo to Senior Support Agent for High path.
9. SLA Check	Optionally check whether the ticket is older than two days.
10. Final Message	Return the corresponding action message.

Priority Automation Logic




Autolaunched Flow
Start
Get Account
Get Records
Store Account Id
Assignment
Get Ticket
Get Records
Store Ticket Id
Assignment
Analyze Description
Decision
High Priority
Medium Priority
Default Outcome
High Path
Assignment
Medium Path
Assignment
Low Path
Assignment
Task
Create Records
Final Medium Message
Assignment
Final Low Message
Assignment
Set Assigned Agent
Assignment
Set Final Message
Assignment
Is High Priority
Decision
High Priority
Default Outcome
End
End



6.6 Agentforce Configuration

The Agentforce configuration uses a subagent named Support Ticket Priority Analysis. Its scope is limited to ticket analysis, priority determination and triggering the backend Flow action. It is instructed not to perform unrelated billing, subscription or account-update tasks.

6.7 Agentforce Action

Property	Configuration
Action	Support Ticket Intelligence Flow action
Input	varAccountName
Loading Text	Analyzing ticket details and determining priority...
Backend	Support_Ticket_Intellegence Auto-Launched Flow
Purpose	Retrieve the latest ticket, analyze priority and return handling outputs

6.8 Input and Output Contract

Direction	Variable	Meaning
Input	varAccountName	Customer Account Name
Output	varAccountId	Account identifier
Output	varTicketId	Analyzed ticket identifier
Output	varPriorityLevel	Calculated priority
Output	varAssignedTo	Assigned support level
Output	varActionMessage	Final handling message

7. FUNCTIONAL & PERFORMANCE TESTING

7.1 Test Strategy

Testing covers the complete automation chain: input collection, Account lookup, ticket retrieval, priority decision, Task creation, assignment output and Agentforce response.

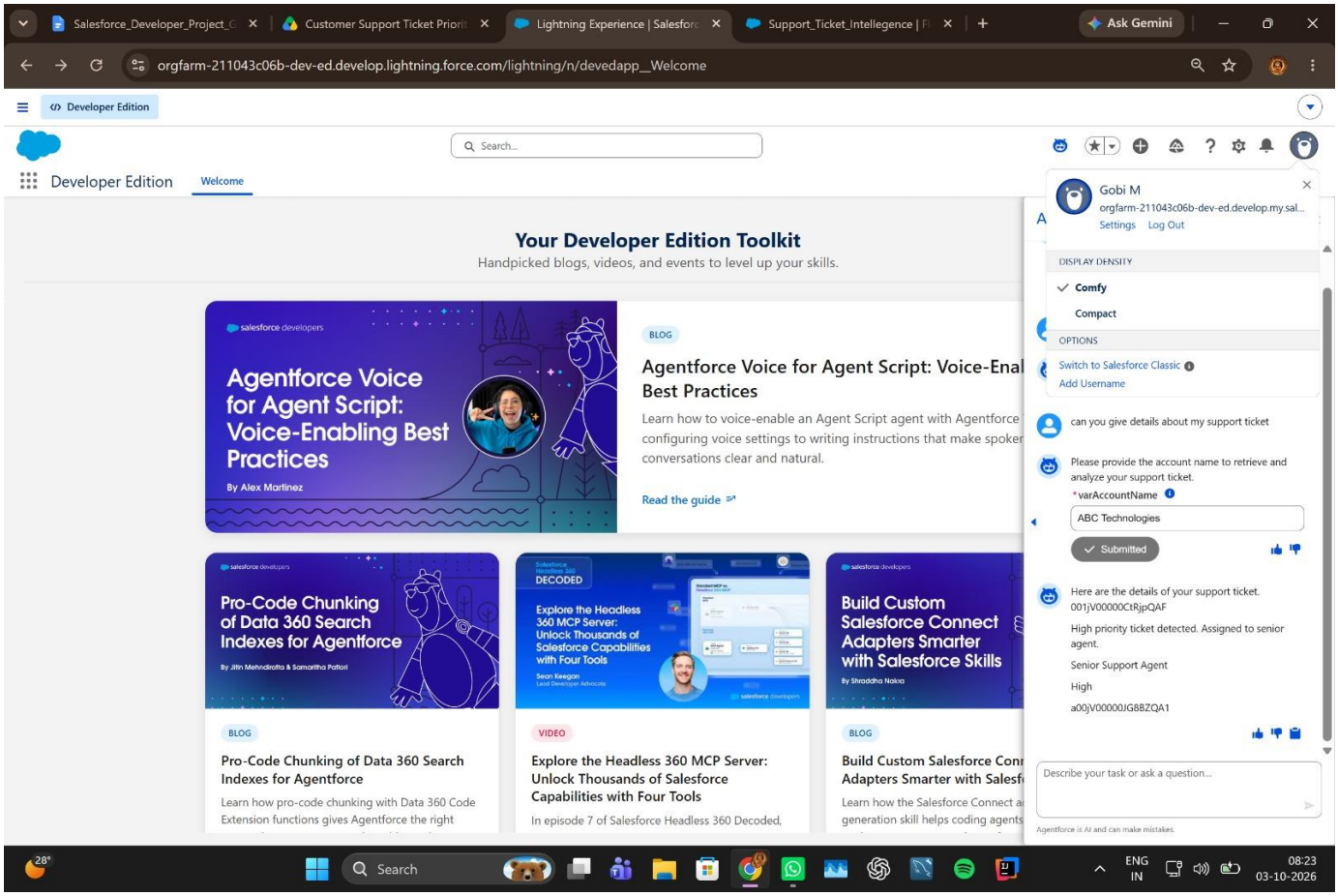
7.2 Test Cases

Test ID	Scenario	Input Condition	Expected Result
TC-01	High priority	Description contains urgent	Priority = High
TC-02	High priority	Description contains not working	Priority = High
TC-03	High priority	Description contains failure	Priority = High1
TC-04	Medium priority	Description contains issue	Priority = Medium
TC-05	Medium priority	Description contains slow or delay	Priority = Medium
TC-06	Low priority	No configured urgency keyword	Priority = Low
TC-07	High task	High priority returned	Urgent Ticket Handling Task created
TC-08	Agentforce	Account Name supplied	Agentforce returns analysis outputs

7.3 High-Priority Task Validation

The High-priority scenario was validated with a ticket description containing multiple High indicators. The resulting Task used the configured subject 'Urgent Ticket Handling', Priority 'High' and Status 'Not Started', and was related to the support ticket through WhatId.

Validation	Observed Configuration
Task Subject	Urgent Ticket Handling
Priority	High
Status	Not Started
Related Record	Support Ticket Intelligence
Assigned Support Level	Senior Support Agent



7.4 Agentforce Validation

A successful Agentforce test should return the ticket identifier, calculated priority, assigned support level and action message after the Flow action executes.

7.5 Test Summary

- Account and latest-ticket retrieval are part of the automated Flow.
- High, Medium and Low outcomes follow the configured decision rules.
- High-priority cases trigger urgent Task creation.
- Agentforce provides the conversational entry point to the automation.

- The implementation is deterministic keyword-based classification rather than a trained ML classifier.

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Reduces manual ticket prioritization.
- Creates a clear High-priority handling path.
- Automates urgent Task creation.
- Provides conversational access through Agentforce.
- Uses Salesforce records as the source of operational data.
- Separates conversational AI from controlled backend Flow actions.
- Can be extended with additional assignment and SLA rules.
- Can be version-controlled as Salesforce DX metadata.

8.2 Disadvantages

- Priority classification depends on configured keywords.
- Keyword matching may not understand complex context or indirect urgency.
- The latest-ticket lookup depends on correct Account association.
- Flow and Agentforce configuration must be maintained as requirements change.
- Current assignment output is simple and can be expanded for multiple teams.
- SLA monitoring is currently limited rather than a full escalation engine.
- Advanced analytics are not part of the current implementation.
- True predictive ML classification would require a future model and training dataset.

9. CONCLUSION

The project demonstrates an end-to-end Salesforce support automation workflow in which Agentforce provides conversational interaction and Salesforce Flow performs controlled backend processing. The system identifies the latest support ticket for an Account, analyzes its description, determines priority, creates urgent work for High-priority cases and returns a structured result. The design provides a practical foundation for expanding toward richer assignment, SLA monitoring, analytics and future ML-based classification.

10. FUTURE SCOPE

- Introduce additional urgency indicators and business rules.
- Add a trained machine-learning priority classifier for true predictive scoring.
- Use historical resolution data to improve priority and SLA-risk prediction.
- Support multiple teams with workload-aware assignment.
- Add escalation notifications and richer SLA monitoring.
- Build dashboards for priority distribution, response time and resolution trends.

- Expand Agentforce outputs with ticket status, issue type and recommended next actions.
- Add confidence scores and explanation fields for future AI-based classification.

11. APPENDIX

A. Custom Object

Support_Ticket_Intelligence__c

B. Key Salesforce Records

- Account
- Contact
- Support Ticket Intelligence
- User
- Task

C. Flow

Support_Ticket_Intellegence – Auto-Launched Flow

D. Agentforce Subagent

Support Ticket Priority Analysis

E. Agentforce Action

Flow-based Support Ticket Intelligence action with Account Name input and ticket-analysis outputs.

F. Original Visuals Included

- Project ecosystem
- Support Agent empathy map
- Solution ideation matrix
- Customer support journey
- System requirements model
- Solution architecture
- Data model
- Priority automation logic
- Agentforce conversation flow
- Development roadmap
- Functional test coverage
- Future enhancement roadmap

THANK YOU

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06/10/2026
Date of issue

Dr. Abhilasha Gaur
CEO
IT-ITeS SSC Nasscom

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Certificate Details

Candidate Name	gobi gobi
Assessment/Course Name	SB - Salesforce Developer
Date of Issue	06/10/2026
Certification ID	FSP/2026/10/10463708
Category Gold>=70% /Silver 60%-69% /Bronze 50%-59%	Gold

FSP/2026/10/10463708

Assessment Score

Module Name/NOS ID	NSQF Level	Maximum Marks	Marks Obtained	Percentage
M001	4	20	20	100.00
M002	4	20	20	100.00
M003	4	20	20	100.00
M004	4	20	20	100.00
M005	4	20	20	100.00
Total		100	100	100.00

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